

Biotechnology and Food Engineering

Process Biotechnology

Spring 2026

Tutorial 10 – scale-up and exam rehearsal - Solutions

Question 1

You are working in fermentation factory trying to develop a procedure to produce antibiotics from fungi. The laboratory experiments are carried in fermenter of 50 liter with engine power of 0.15 HP. The experiments reveal that the most important factor to obtain high antibiotics concentration is shear stress that enables to keep a small pellet of the fungus mycelium. The factory aims to build production reactor in volume of 200,000 liter.

In your opinion, what should be the engine power in the production reactor?

Solution:

<u>Lab</u>	<u>Factory</u>
$V_1 = 50 \text{ lit}$	$V_2 = 200,000 \text{ lit}$
$P_1 = 0.15 \text{ HP}$	$P_2 = ?$

The important factor is the shear stress.

When keeping high turbulent flow and same type of stirrer, N_p is constant:

$$P = N_p \cdot N^3 \cdot D_i^5 \cdot \rho \rightarrow P \propto N^3 D_i^5$$

$$S = \pi \cdot D_i \cdot N \rightarrow S \propto D_i \cdot N$$

When keeping the $\frac{h}{D}$ and $\frac{D_i}{D}$ ratios constant:

$$V = \frac{\pi \cdot D^3 \cdot C}{4} \rightarrow V \propto D^3 \propto D_i^3$$

We want to find connection between D_{i_1} and D_{i_2} , and connection between N_2 to N_1 .

$$\frac{P_2}{P_1} = \frac{N_2^3 D_{i_2}^5}{N_1^3 D_{i_1}^5}$$

From the volume data:

$$\frac{V_2}{V_1} = \frac{D_{i_2}^3}{D_{i_1}^3} \rightarrow \frac{D_{i_2}}{D_{i_1}} = \sqrt[3]{\frac{V_2}{V_1}}$$

$$\frac{D_{i_2}}{D_{i_1}} = \sqrt[3]{\frac{200,000 \text{ lit}}{50 \text{ lit}}} = 15.874$$

We are interested to maintain the shear stress, therefore:

$$D_{i_1} \cdot N_1 = D_{i_2} \cdot N_2$$

$$\frac{N_2}{N_1} = \frac{D_{i_1}}{D_{i_2}} = \frac{1}{15.874} = 0.063$$

Place it into the engine powers ratio:

$$\frac{P_2}{P_1} = (0.063)^3 \cdot (15.874)^5 = 251.9$$

$$P_2 = 251.9 \cdot 0.15 \text{ HP}$$

$$P_2 = \mathbf{37.8 \text{ HP}}$$

Question 2

A factory grows bacteria in batch 200-liter fermenter, for the purpose of biomass production. The bacteria is inoculated into the fermenter at initial concentration of 80 ppm, and the fermentation was designed so that each fermentation cycle will obtain 8 kg of bacteria. The bacteria grows on molasses which consists of 56% solids (87% of it are sugars), and its initial concentration in the medium is 300 gr molasses per liter medium.

Regular air, which includes $6300 \frac{\text{spores}}{\text{lit}_{\text{air}}}$ after initial filtration, is supplied into the fermenter at rate of 1.6 VVM, after passing through continuous sterilizer with volume of 1.2 m^3 . The sterilization carried at temperature of 145°C , and the sterilization constant at this temperature is 6.85 min^{-1} .

Additional data: the bacteria start immediately logarithmic growth with doubling time of 80 minutes. The growth yield of the bacteria on the molasses sugars is $0.57 \frac{gX}{gS}$.

A. What is the probability for contamination in the fermentation (expressed as 1 from X fermentation)?

In order to produce larger amounts of biomass, the factory management bought new fermenter with volume of 13 m^3 , with the same type of stirrer but the new stirrer diameter is bigger by 4-fold than the stirrer in the old fermenter. The flow regime in both old and new fermenters during fermentation is kept turbulent, and from experiments done in the Pilot Plant, it was found that the most important factor needed to be kept for proper growth of the bacteria is shear stress.

B. In the old fermenter they used an engine with the power of 0.23 horsepower (HP). What is the engine power required in the new fermenter?

Solution:

Given:

<u>Fermentation</u>	<u>Molasses</u>	<u>Air</u>
$V = 200 \text{ lit}$	$M_0 = 300 \frac{gM}{lit}$	$N'_0 = 6.3 \cdot 10^3 \frac{\text{spore}}{\text{lit}_{\text{air}}}$
$X_0 = 80 \text{ ppm} = 0.08 \frac{gX}{lit}$	$\frac{gSolid}{gM} = 0.56$	$\left(\frac{Q_{\text{air}}}{V}\right)_{in} = 1.6 \frac{\text{lit}_{\text{air}}}{lit \cdot hr}$
$W_{x_f} = 8 \text{ kgX}$	$\frac{gS}{gSolid} = 0.87$	$V_{\text{sterilizer}} = 1.2 \text{ m}^3$
$Y_{\frac{X}{S}} = 0.57 \frac{gX}{gS}$	$\downarrow$	$K_d = 6.85 \frac{1}{min}$
$t_d = 80 \text{ min} = 1.33 \text{ hr}$	$S_0 = 300 \frac{gM}{lit} \cdot 0.56 \cdot 0.87$	
$\downarrow$	$= 146.16 \frac{gS}{lit}$	
$\mu = \frac{\ln 2}{t_d} = 0.521 \frac{1}{hr}$		

A. $N_f = ?$

The number of spores enter the sterilizer during the fermentation can be found from the growth period:

$$N_0 = N'_0 \cdot \frac{Q}{V_{in}} \cdot V \cdot \Delta t_{log}$$

It is given that the bacteria immediately start logarithmic growth, so we can find the growth period from the logarithmic growth equation:

$$\ln\left(\frac{X_f}{X_0}\right) = \mu \cdot \Delta t_{log}$$

$$X_f = \frac{W_{X_f}}{V} = \frac{8000 \text{ gX}}{200 \text{ lit}} = 40 \frac{\text{gX}}{\text{lit}}$$

$$\Delta t_{log} = \frac{\ln\left(\frac{X_f}{X_0}\right)}{\mu} = \frac{\ln\left(\frac{40 \frac{\text{gX}}{\text{lit}}}{0.08 \frac{\text{gX}}{\text{lit}}}\right)}{0.521 \frac{1}{\text{hr}}} = 11.93 \text{ hr}$$

Therefore:

$$N_0 = 6.3 \cdot 10^3 \frac{\text{spore}}{\text{lit}_{air}} \cdot 1.6 \frac{\text{lit}_{air}}{\text{lit} \cdot \text{min}} \cdot 200 \text{ lit} \cdot 11.93 \text{ hr} \cdot 60 \frac{\text{min}}{\text{hr}} = 1.44 \cdot 10^9 \frac{\text{spore}}{\text{ferm}}$$

We assume that the heating and cooling time of the air is negligible, so $t_{staying} = t_{st}$.

The staying time can be found from the air flow rate and the sterilizer volume:

$$Q_{air} = \frac{Q_{air}}{V_{in}} \cdot V = 1.6 \frac{\text{lit}_{air}}{\text{lit} \cdot \text{min}} \cdot 200 \text{ lit} = 320 \frac{\text{lit}_{air}}{\text{min}}$$

$$t_{staying} = \frac{V_{sterilizer}}{Q_{air}} = \frac{1200 \text{ lit}}{320 \frac{\text{lit}}{\text{min}}} = 3.75 \text{ min}$$

From the sterilization equation, assuming that the air is heated to its sterilization temperature very quickly (meaning: $t_{st} = t_{staying}$):

$$\ln\left(\frac{N_f}{N_0}\right) = -K_d \cdot t_{st}$$

$$N_f = N_0 \cdot e^{-K_d \cdot t_{st}} = 1.44 \cdot 10^9 \frac{\text{spore}}{\text{ferm}} \cdot e^{-6.85 \frac{1}{\text{min}} \cdot 3.75 \text{ min}}$$

$$N_f = 0.01 \frac{\text{spore}}{\text{ferm}}$$

Meaning, the probability for contamination is 1 out of 100 fermentations.

B. $P_{new} = ?$

Given: $P_{old} = 0.23 \text{ HP}$

From the N_p equation: $P = N_p \cdot \rho \cdot N^3 \cdot D_i^5$

It is said in the question that the most important parameter for a proper growth is shear stress, thus, we aspire to maintain it, meaning: $S_{new} = S_{old}$, while $S = \pi \cdot N \cdot D_i$.

Therefore, in order to maintain the same shear stress:

$$\pi \cdot N_{new} \cdot D_{i_{new}} = \pi \cdot N_{old} \cdot D_{i_{old}}$$

It is given that $D_{i_{new}} = 4D_{i_{old}}$, therefore:

$$\pi \cdot N_{new} \cdot 4D_{i_{old}} = \pi \cdot N_{old} \cdot D_{i_{old}}$$

$$N_{new} = \frac{N_{old}}{4}$$

As the flow regime will be kept turbulent, and the stirrer is from the same type - N_p stays the same. Hence,

$$\frac{P_{new}}{P_{old}} = \frac{N_p \cdot \rho \cdot \left(\frac{N_{old}}{4}\right)^3 \cdot (4D_{i_{old}})^5}{N_p \cdot \rho \cdot (N_{old})^3 \cdot (D_{i_{old}})^5} = 4^2 = 16$$

$$P_{new} = 16 \cdot P_{old} = 16 \cdot 0.23 \text{ HP}$$

$$P_{new} = 3.68 \text{ HP}$$

The engine power required in the new fermenter is 3.68 horsepower.

Question 3

Continuous yeast fermentation with circulation and concentration is performed in fermenter of 500 liter, for the purpose of extracellular product production. The circulation flow rate in the fermenter is 40% of the fresh feed flow rate, and the concentration factor in the continuous cell separator is 2.5. The protein production yield is $0.46 \frac{gP}{gX}$.

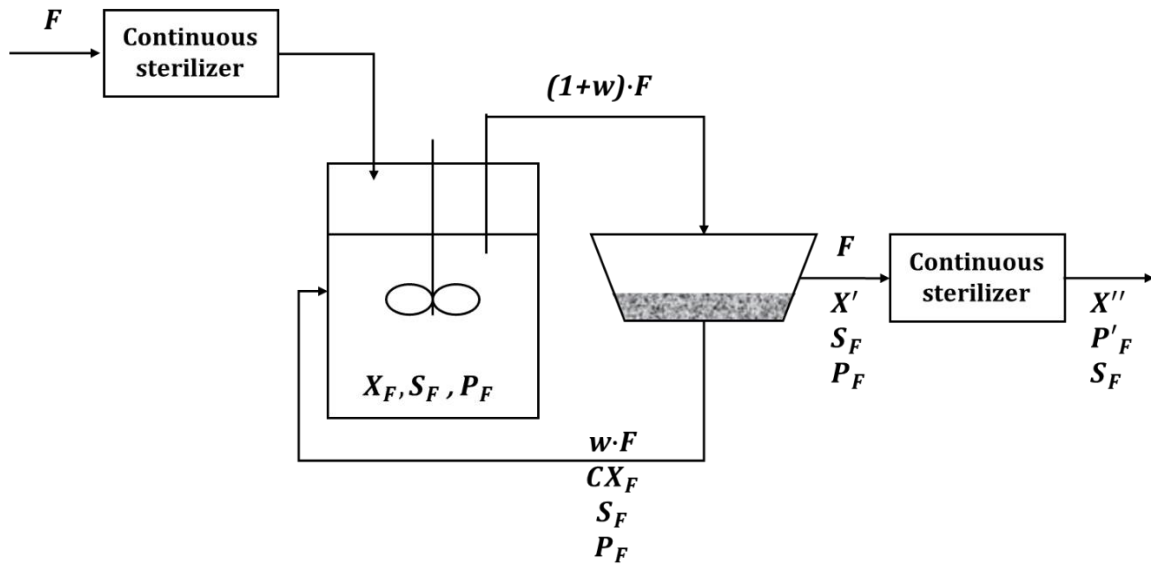
The fresh feed that enters the fermenter is being sterilized at a temperature of 142°C , for sterilization period of 10 minutes. The contaminating spores in the fresh feed, before its sterilization, is $1 \cdot 10^8 \frac{\text{spores}}{\text{lit}}$, and the death constant of the spores in the given temperature is 3.25 min^{-1} . The sterilization was planned so that in one continuous working year, no more than one contaminating spore will enter the fermenter.

The fermenter is continuously supplied with normal air (21% oxygen), filtered and sterilized, at pressure of 1.4 atmospheres. In the fermentation conditions, it is known that Henry's coefficient is $1.47 \frac{\text{atm} \cdot \text{lit}}{\text{mmol } O_2}$, the oxygen transfer rate coefficient is 1100 per hour, and during steady state the oxygen electrode read constantly 15%. The yeast yield on oxygen is $0.6 \frac{gX}{gO_2}$. The medium exiting the fermenter is also streaming through continuous sterilization, before it continues to separation processes for isolation of the produced protein. This sterilization aims to ensure that the yeast concentration in the liquid that enters the separation processes will not exceed 1 ppm. The sterilization of the liquid exiting the fermenter is done in temperature of 123°C , and the death constant of the yeast in this temperature is 1.2 min^{-1} .

The protein produced in this fermentation is partially decomposed in heat due the sterilization of the stream exiting the fermenter, while its decomposition constant in temperature of 123°C is 100-fold less than the death constant of the yeast in this temperature.

What is the protein concentration ($\frac{g}{\text{lit}}$) in the sterilized flow that continues to the separation processes?

Solution:



Fermentation

$$V = 500 \text{ lit}$$

$$w = 0.4$$

$$C = 2.5$$

$$Y_{\frac{P}{X}} = 0.46 \frac{gP}{gX}$$

Sterilization

$$T_{st} = 142^\circ\text{C}$$

$$t_{st} = 10 \text{ min}$$

$$N_0 = 1 \cdot 10^8 \frac{\text{spores}}{\text{lit}}$$

$$K_d = 3.25 \frac{1}{\text{min}}$$

$$N'_F = \frac{1 \text{ spore}}{1 \text{ year}}$$

Air

$$\%O_2 = 21\% = 0.21$$

$$P = 1.4 \text{ atm}$$

$$H = 1.47 \frac{\text{atm} \cdot \text{lit}}{\text{mmol}O_2}$$

$$K_L a = 1100 \frac{1}{\text{hr}}$$

$$C_L = 0.15 \cdot C^*$$

$$Y_{\frac{X}{O}} = 0.6 \frac{gX}{gO_2}$$

Exit sterilization

$$N_F = 1 \cdot 10^{-3} \frac{gX}{\text{lit}}$$

$$T_{st} = 123^\circ\text{C}$$

$$K_d = 1.2 \frac{1}{\text{min}}$$

$$K' = \frac{K_d}{100} = 0.012 \frac{1}{\text{min}}$$

$$P'_F = ?$$

The decomposition equation:

$$\ln\left(\frac{P'_F}{P_F}\right) = -K' \cdot \Delta t$$

We need to find the sterilization period (Δt) and the concentration too (P_F).

The sterilization period we can find from the yeast sterilization. For that we need X' .

P_F can be calculated from balances on the fermenter.

X' can be calculated from biomass balance on the centrifuge.

For the fermentation $\rightarrow$ standard circulation and concentration:

$$\mu = D \cdot [1 + w \cdot (1 - C)]$$

$$X_F = \frac{\frac{Y_X}{S} \cdot (S_0 - S_F)}{[1 + w \cdot (1 - C)]}$$

$$P_F = \frac{Y_P}{X} \cdot X_F \cdot [1 + w \cdot (1 - C)] = \frac{Y_P}{S} \cdot (S_0 - S_F)$$

There is no information on the substrate concentration, so we can't use it. We should pay attention that we can use the information given on the air:

$$OTR = OUR$$

$$K_L a \cdot (C^* - C_L) = \frac{\mu \cdot X_F}{\frac{Y_X}{O}} \rightarrow X_F = \frac{K_L a \cdot (C^* - C_L) \cdot Y_X \cdot \frac{O}{O}}{\mu}$$

Calculate C^* using Henry's law:

$$C^* = \frac{P \cdot X_{O_2}}{H} = \frac{1.4 \text{ atm} \cdot 0.21}{1.47 \frac{\text{atm} \cdot \text{lit}}{\text{mmol } O_2} \cdot \frac{1 \text{ mol}}{32 \text{ g}} \cdot 10^{-3} \frac{\text{mmol}}{\text{mol}}} = 6.4 \cdot 10^{-3} \frac{\text{g } O_2}{\text{lit}}$$

We still have to find μ and X_F .

We can find μ through D . To do that, we need to find F .

We will look on the sterilization process of the incoming medium:

$$\ln\left(\frac{N_F}{N_0}\right) = -K_d \cdot t_{st}$$

$$\ln\left(\frac{N_F}{1 \cdot 10^8 \frac{\text{spores}}{\text{lit}}}\right) = -3.25 \frac{1}{\text{min}} \cdot 10 \text{ min}$$

$$N_F = 7.68 \cdot 10^{-7} \frac{\text{spores}}{\text{lit}}$$

It is given that the probability for contamination is one spore per year. We can write it as:

$$(Q_{\text{spores}}) \cdot 1 \text{ year} = 1 \text{ spore}$$

Find the spore flow rate (Q_{spores}):

$$Q_{\text{spores}} = F \cdot N_F$$

$$\left(F \cdot 7.68 \cdot 10^{-7} \frac{\text{spores}}{\text{lit}}\right) \cdot 1 \text{ year} \cdot 365 \frac{\text{day}}{\text{year}} \cdot 24 \frac{\text{hr}}{\text{day}} = 1 \text{ spore}$$

$$F = 148.6 \frac{\text{lit}}{\text{hr}}$$

$$D = \frac{F}{V} = \frac{148.6 \frac{\text{lit}}{\text{hr}}}{500 \text{ lit}} = 0.297 \frac{1}{\text{hr}}$$

$$\mu = 0.297 \frac{1}{\text{hr}} \cdot [1 + 0.4 \cdot (1 - 2.5)] = 0.1188 \frac{1}{\text{hr}}$$

Now let's find X_F :

$$X_F = \frac{K_L a \cdot C^* (1 - C_L) \cdot Y_{X/\bar{O}}}{\mu} = \frac{1100 \frac{1}{\text{hr}} \cdot 6.4 \cdot 10^{-3} \frac{\text{gO}_2}{\text{lit}} \cdot (1 - 0.15) \cdot 0.6 \frac{\text{gX}}{\text{gO}_2}}{0.1188 \frac{1}{\text{hr}}}$$

$$X_F = 30.22 \frac{\text{gX}}{\text{lit}}$$

$$P_F = Y_{P/\bar{X}} \cdot X_F \cdot [1 + w \cdot (1 - C)] = 0.46 \frac{\text{gP}}{\text{gX}} \cdot 30.22 \frac{\text{gX}}{\text{lit}} [1 + 0.4 \cdot (1 - 2.5)] = 5.56 \frac{\text{gP}}{\text{lit}}$$

To find X' we will perform balance on the centrifuge:

$$(1 + w) \cdot \cancel{F} \cdot X_F = w \cdot \cancel{F} \cdot C \cdot X_F + \cancel{F} \cdot X'$$

$$X' = X_F \cdot [1 + w \cdot (1 - C)] = 30.22 \frac{gX}{lit} \cdot [1 + 0.4 \cdot (1 - 2.5)] = 12.1 \frac{gX}{lit}$$

Place it in the sterilization equation to find the second sterilization period:

$$\ln\left(\frac{X''}{X'}\right) = -K_d \cdot \Delta t$$

$$\ln\left(\frac{1 \cdot 10^{-3} \frac{gX}{lit}}{12.1 \frac{gX}{lit}}\right) = -1.2 \frac{1}{min} \cdot \Delta t$$

$$t = 7.83 \text{ min}$$

Place it in the decomposition equation:

$$\ln\left(\frac{P'_F}{5.56 \frac{gP}{lit}}\right) = -0.012 \frac{1}{min} \cdot 7.83 \text{ min}$$

$$P'_F = 5.06 \frac{gP}{lit}$$

Question 4

A factory performs batch fermentation of bacteria that grow on medium contains two different sugars types. The 1.5 m³ fermenter is initially inoculated with 75 grams of dry bacteria. The bacteria stay in lag stage for 1.5 hour, followed by logarithmic growth on the first sugar (marked as “substrate A”). Once substrate A is completed, the bacteria instantly continue to grow logarithmic on the second sugar (“substrate B”), for 4 additional hours, but at different growth rate. The initial concentration of substrate B in the fermenter was 2%, the growth yield of the bacteria on this substrate is 0.24 $\frac{gX}{gS_B}$, while the growth on this substrate ends when its concentration in the fermenter is 250 ppm. At this point, the fermenter is discharged, and its content continues to separation processes aiming to separate a specific intercellular protein that constitute 39% of the bacteria biomass. The time period for discharging the fermenter, cleaning and prepare it for the next fermentation takes 3.5 hours. The productivity (volumetric production rate) of this process is 0.171 $\frac{gP}{lit \cdot hr}$. The

factory works constantly, 30 days a month, and performs 40 fermentation cycles per month. After the separation processes, which their total yield is 97.6%, the monthly output of the factory is 180 kg of clean and purified protein.

Regular air (21% oxygen) is supplied into the fermenter at a pressure of 1.3 atmospheres and temperature of 37°C, and in these fermentation conditions the maximal solubility of the air in the fermentation medium is 4.85%. The oxygen yield on the bacterial biomass when it grows on substrate A is $1.385 \frac{gO_2}{gX}$, and the biomass yield on oxygen when it grows on substrate B is $1.2 \frac{gX}{gO_2}$. The oxygen concentration dissolved in saturation is 8 ppm.

What is the minimal required oxygen transfer rate ($K_L a$), in order to ensure that the oxygen will not be the limiting factor?

Solution:

<u>Fermentation</u>
$V = 1500 \text{ lit}$
$W_{X_0} = 75 \text{ gX} \rightarrow X_0 = \frac{W_{X_0}}{V} = 0.05 \frac{\text{gX}}{\text{lit}}$
$t_{lag} = 1.5 \text{ hr}$
$t_{logB} = 4 \text{ hr}$
$S_{B_0} = 20 \frac{\text{gS}}{\text{lit}}$
$Y_{\frac{X}{S_B}} = 0.24 \frac{\text{gX}}{\text{gS}_B}$
$S_{B_F} = 250 \text{ ppm} = 0.25 \frac{\text{gS}}{\text{lit}}$
$Y_{\frac{P}{X}} = 0.39 \frac{\text{gP}}{\text{gX}}$
$t_{maintenance} = 3.5 \text{ hr}$
$P_{rod} = 0.171 \frac{\text{gP}}{\text{lit} \cdot \text{hr}}$
$30 \frac{\text{day}}{\text{month}}, 40 \frac{\text{fermentation}}{\text{month}}$
$\eta = 97.6\% = 0.976$

<u>Air</u>
$\%O_2 = 21\% = 0.21$
$P = 1.3 \text{ atm}$
$T = 37^\circ\text{C}$
$\% \text{ solubility}_{(max.)} = 4.85\%$
$Y_{\frac{O}{X_A}} = 1.385 \frac{\text{gO}_2}{\text{gX}}$
$Y_{\frac{X_B}{O}} = 1.2 \frac{\text{gX}}{\text{gO}_2}$
$C^* = 8 \text{ ppm} = 0.008 \frac{\text{gO}_2}{\text{lit}}$

$$W_{P_{clean}} = 180 \frac{kg}{month}$$

$$K_L a = ?$$

So that the oxygen will not be limiting factor:

$$OUR_{max} = OTR_{max}$$

Check when OUR is maximal – two options: at the end of the fermentation or at the end of the growth on substrate A.

$$OUR = \frac{\mu \cdot X_F}{Y_{\frac{X}{O}}}$$

In order to find OUR in both cases, we need to find μ and X_F in each of these ending points.

Find X_F from the data on the product:

$$W_P = \frac{180 \frac{kgP}{month}}{40 \frac{ferm}{month}} \cdot \frac{1}{0.976} = 4.61 \frac{kgP}{ferm}$$

$$P_F = \frac{4610 \frac{gP}{lit}}{1500 \frac{lit}{lit}} = 3.07 \frac{gP}{lit}$$

$$Y_{\frac{P}{X}} = \frac{(P_F - P_0)}{(X_F - X_0)}$$

As the product is intracellular and part of the microorganism biomass,

$$P_0 = Y_{\frac{P}{X}} \cdot X_0 = 0.39 \frac{gP}{gX} \cdot 0.05 \frac{gX}{lit} = 0.0195 \frac{gP}{lit}$$

$$0.39 \frac{gP}{gX} = \frac{3.07 \frac{gP}{lit} - 0.0195 \frac{gP}{lit}}{X_F - 0.05 \frac{gX}{lit}}$$

$$X_F = 7.87 \frac{gX}{lit}$$

X_A we can find from the yield on substrate B:

$$Y_{\frac{X}{S_B}} = \frac{(X_F - X_A)}{(S_{B_0} - S_{B_F})}$$

$$0.24 \frac{gX}{gS_B} = \frac{\left(7.87 \frac{gX}{lit} - X_A\right)}{(20 - 0.25) \frac{gS}{lit}}$$

$$X_A = 3.13 \frac{gX}{lit}$$

Now we only need to find the growth rates. We can use the growth equation on substrate B:

$$\ln\left(\frac{X_F}{X_A}\right) = \mu_B \cdot t_{log_B}$$

$$\ln\left(\frac{7.87 \frac{gX}{lit}}{3.13 \frac{gX}{lit}}\right) = \mu_B \cdot 4 \text{ hr}$$

$$\mu_B = 0.23 \frac{1}{hr}$$

For A, we need to find the logarithmic growth period (t_{log_A}). We can find it from the total time:

$$P_{rod} = \frac{\Delta P}{t_T} = \frac{(3.07 - 0.0195) \frac{gP}{lit}}{t_T} = 0.171 \frac{gP}{lit \cdot hr}$$

$$t_T = 17.84 \text{ hr}$$

$$t_T = t_{lag} + t_{log_A} + t_{log_B} + t_{maintenance}$$

$$17.84 \text{ hr} = 1.5 \text{ hr} + t_{log_A} + 4 \text{ hr} + 3.5 \text{ hr}$$

$$t_{logA} = 8.84 \text{ hr}$$

Place it in the growth equation:

$$\ln\left(\frac{3.13 \frac{gX}{lit}}{0.05 \frac{gX}{lit}}\right) = \mu_A \cdot 8.84 \text{ hr}$$

$$\mu_A = 0.468 \frac{1}{hr}$$

Now we can calculate the OUR:

$$OUR_{A_{max}} = \frac{0.468 \frac{1}{hr} \cdot 3.13 \frac{gX}{lit}}{\frac{1}{1.385 \frac{gO}{gX}}} = 2.03 \frac{gO_2}{lit \cdot hr}$$

$$OUR_{B_{max}} = \frac{0.23 \frac{1}{hr} \cdot 7.87 \frac{gX}{lit}}{1.2 \frac{gO}{gX}} \cong 1.51 \frac{gO_2}{lit \cdot hr}$$

Therefore, OUR_{max} is at the end of the growth on substrate A.

$$OUR_{max} = 2.03 \frac{gO_2}{lit \cdot hr}$$

The minimal required $K_L a$ is received when $C_L = 0$. At that point:

$$K_L a_{min} \cdot C^* = 2.03 \frac{gO_2}{lit \cdot hr}$$

$$K_L a_{min} = \frac{2.03 \frac{gO_2}{lit \cdot hr}}{0.008 \frac{gO_2}{lit}}$$

$$K_L a_{min} = 253 \frac{1}{hr}$$